

Kita Malaysia

An Interactive 3D Malaysian Cultural & Heritage Explorer

Project Executive Summary

Kita Malaysia is a modern, high-performance web experience crafted to celebrate Malaysia's rich tapestry of cultures, heritage, and national unity. Built with Next.js 16, React 19, TypeScript, and Three.js, it unites Peninsular Malaysia and Malaysian Borneo through an interactive 3D miniature landscape, authentic cultural stories, an educational trivia quiz, and a gamified explorer passport. The application emphasizes accessibility, digital sovereignty (zero backend/API dependence), and deep reverence for Malaysia's living heritage.

DEVELOPER

Haziq Farhan (hzqfarhan)

GITHUB REPOSITORY

github.com/hzqfarhan/kitamalaysia

COMPETITION TRACK

Web Application / Cultural Heritage

SUBMISSION DATE

September 2026

01. Merdeka Theme & National Concept

"Made of Many. Moving as One."

The Vision: Independence Through Mutual Respect & Unity

Independence (*Merdeka*) is more than a historical milestone achieved on August 31, 1957; it is a living commitment to mutual harmony, mutual understanding, and collective resilience. True national identity is forged not by erasing differences, but by celebrating how our diverse traditions, geography, and communities strengthen one another.

Kita Malaysia embodies this ethos by bringing Malaysian history and living traditions to the fingertips of the youth. Rather than presenting static textbook paragraphs, the website translates national heritage into an explorative, interactive spatial canvas.

Six Cultural Pillars Across East & West Malaysia

1. Kuala Lumpur

INNOVATION

Rooted in culture, reaching for the sky. Showcases the PETRONAS Twin Towers, highlighting how modern architecture is grounded in Islamic geometric design (the eight-pointed star).

2. George Town (Penang)

UNESCO HERITAGE

Streets of living exchange. Celebrates centuries of multicultural harmony, historic shophouses, and shared community spaces of the Straits of Malacca.

3. Melaka

LIVING HERITAGE

The historic maritime crossroads. Traces the historical meeting of Malay, Portuguese, Dutch, and Peranakan cultures that laid the foundation of Malaysian pluralism.

4. Kelantan

ART & CRAFT

Craftsmanship taking flight. Spotlights the traditional *Wau Bulan* kite, symbolizing patience, craftsmanship, and soaring national ambition.

5. Mount Kinabalu (Sabah)

NATURE & ECOLOGY

A sacred ecosystem. Highlights Sabah's UNESCO World Heritage national park, its sacred cultural reverence by indigenous communities, and remarkable biodiversity.

6. Sarawak

COMMUNITY UNITY

"Different doors. A shared space." Explores the philosophy of the indigenous longhouse — a profound architectural metaphor for national unity and harmony.

Educational Authenticity: All stories, historical milestones, and cultural facts are cross-referenced with accredited sources including the *UNESCO World Heritage Centre*, *Tourism Malaysia*, and the *Sarawak Tourism Board*.

Core Technical Features

- **Interactive Three.js 3D Miniature Landscape:** A low-poly illustrative 3D canvas featuring procedural terrain, customizable orbital camera controls (drag-to-orbit, scroll-to-zoom, auto-reset), animated cloud layers, and a flying *Wau Bulan* kite rendered in real-time WebGL.
- **Interactive Landmark Raycasting:** Users interact with custom 3D pins representing historical regions, triggering synchronized story dialogs and geographic focus.
- **Gamified Explorer Passport:** Six digital collectible passport stamps earned as visitors explore each state's story. Progress and high scores are preserved across browser sessions via `localStorage` with zero account barrier.
- **Malaysia Trivia Quiz Engine:** A 5-question trivia experience testing insights gained across the explorer, featuring explanatory feedback and persistent high score tracking.
- **Accessibility First Design:** Built-in skip links, native accessible dialogs (`<dialog>`), high-contrast typography, and `prefers-reduced-motion` overrides.

Technology Stack Breakdown

LAYER	TECHNOLOGY	PURPOSE & JUSTIFICATION
Core Framework	Next.js 16 (React 19)	App Router with output: 'export' static generation for ultra-fast, edge-ready delivery.
Type Safety	TypeScript 5.9	Strict typing across destination models, quiz structures, and WebGL state machines.
3D WebGL Engine	Three.js 0.186	Lightweight custom 3D geometry without heavy external 3D asset loaders, ensuring sub-second initial loads.
Iconography	Lucide React	Clean, accessible, tree-shaken SVG icon system.
Styling System	Modern CSS3 Variables	Glassmorphism, fluid typography, dark-accented cultural palettes, and responsive layouts.
Data Storage	Client LocalStorage	Zero tracking, zero cookies, privacy-preserving client-side passport & score persistence.

Performance & Production Optimization

The entire application compiles to a purely static bundle in `out/`. By utilizing procedural Three.js geometries for the landscape and optimizing photography, the application achieves lightning-fast first contentful paint (FCP), zero server-side latency, and 100% offline-ready reliability.

03. Educational Impact & Submission Summary

Evaluation
Criteria

Alignment With Competition Criteria

CRITERION	IMPLEMENTATION IN KITA MALAYSIA
Creativity & Design	Playful, modern 3D low-poly miniature aesthetic. Blends cutting-edge WebGL graphics with warm Malaysian cultural motifs and glassmorphism.
Merdeka & Culture Theme	Deep narrative integration of unity, cultural craftsmanship (Wau Bulan), architectural symbolism (Petronas Towers), and communal coexistence (Sarawak longhouses).
Technical Execution	Flawless TypeScript compilation, Next.js 16 static export, real-time 3D camera controls, and accessible dialogs.
User Engagement	Gamified Explorer Passport and Malaysia Trivia Quiz turn passive reading into an active, delightful discovery experience.
Digital Ethics & Privacy	Requires no logins, no personal data harvesting, and fully transparent open-source attributions and licensing.

Project Deliverables & Repository Access

Source Code Repository: <https://github.com/hzqfarhan/kitamalaysia>

Branch: main (Clean history, production-ready)

License & Rights: MIT Open Source / Educational Use

"Made of many. Moving as one. Code. Create. Celebrate Malaysia." — Merdeka WebForge 2026